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Conversational FPGA Design Space Exploration: A ReAct-Based LLM Agent Coupled with a Transformer-Based Surrogate Model

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OVERVIEW

Mapping AI models onto FPGAs involves thousands of possible designs. High-Level Synthesis (HLS) evaluates resource use and performance, but each run can take minutes to hours, making exhaustive exploration impractical. This work combines a Transformer surrogate and a ReAct LLM agent to accelerate FPGA design-space exploration by predicting hardware metrics and searching for designs under user-defined constraints.

One AI model
Many FPGA designs

×

Each design requires
evaluation
Minutes to hours per evaluation

=

Exploration time
bottleneck

Background motivation

Motivated by the wa-hls4ml benchmark [1], which paired hundreds of thousands of real HLS4ML designs with their hardware reports and showed this kind of data could go beyond single-candidate prediction, toward an assistant that actively searches the design space.

One AI model can be mapped onto an FPGA in thousands of ways. HLS translates each configuration into resource and latency estimates, but each run can take minutes to hours, making exhaustive evaluation impractical [2].

This work investigates whether a surrogate model and an LLM agent can jointly support FPGA design-space exploration. Early results show accurate predictions and grounded, constraint-aware exploration.

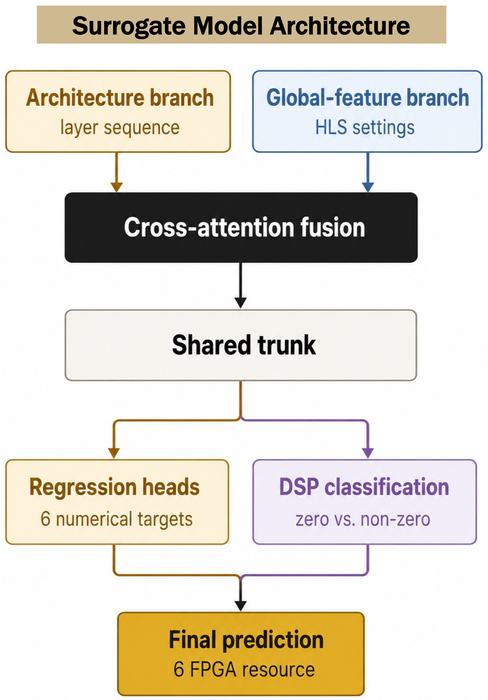
Prediction and search, combined into one system

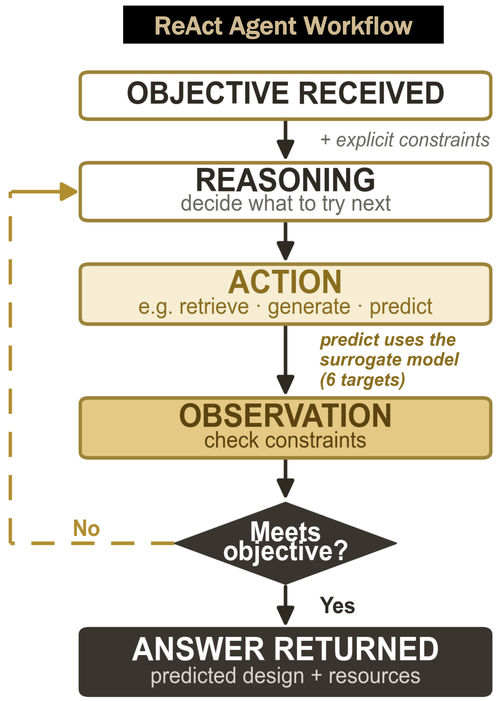
PREDICTION MODEL

The wa-hls4ml dataset connects each FPGA design architecture with its HLS resource and latency report, providing six prediction targets: BRAM, DSP, FF, LUT, cycles, and initiation interval. After cleaning, 619,601 designs were divided into 70% training, 15% validation, and 15% testing. The 2.3-million-parameter surrogate uses six regression heads and a dedicated classifier for zero-valued DSP cases. It was trained for 60 epochs with AdamW, reaching its best validation loss of 0.0126 at epoch 54.

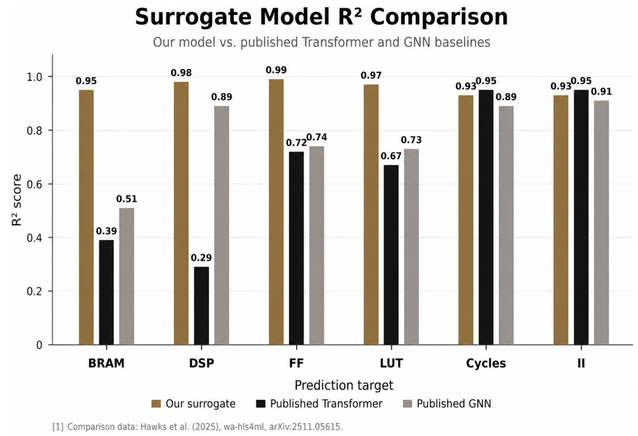
LLM AGENT

Built on the ReAct pattern, the LLM agent transforms user goals into an iterative design search. It reasons about what to try, retrieves or generates a candidate, uses the surrogate to predict six hardware metrics, and observes whether the constraints are satisfied. If they are not, the agent revises the candidate and repeats the cycle. Once the objective is met, it returns the selected design and its predicted resources, without running full HLS at every iteration.





Results



Results and discussion

The surrogate achieved a mean R^2 of 0.96 across six targets on unseen designs, outperforming published baselines on BRAM, DSP, FF, and LUT while remaining competitive on cycles and initiation interval. In the system evaluation, agent grounding reached 100% across all design types, while constraint satisfaction reached 100% for Conv1D and Conv2D and 50% for Dense; fidelity and plausibility varied. Together, the surrogate and ReAct agent demonstrate the potential for grounded, constraint-aware FPGA design-space exploration, although performance remains architecture-dependent. Future work will confirm candidates through HLS, expand the benchmark, improve candidate generation, and compare exploration time against conventional HLS-based approaches.

[1],[awks, B., Weitz, J., Demler, D., et al. (2025). *wa-hls4ml: A benchmark and surrogate models for hls4ml resource and latency estimation*. arXiv:2511.05615.
[2], Bi, J., Schütze, L., & Castrillon, J. (2026). *MING: An automated CNN-to-edge MLIR HLS framework*. arXiv:2602.11966.